

Deep Learning-Based Railway Foreign Object Intrusion Intelligent Perception Using Attention-Aggregated Semantic Segmentation

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Abstract—Foreign object intrusion detection (FOID) is one of the critical tasks to ensure the safe and efficient operation of trains. Semantic segmentation, which involves pixel-level recognition of images, has been widely studied in automatic driving obstacle avoidance. However, unlike road transportation, the operation speed of trains requires higher detection efficiency. The availability of mature railway scenario datasets is limited compared to road transportation datasets. Therefore, considering the complexity of operating scenarios with diverse and unpredictable foreign objects, this article proposes a boundary-assisted dual-branch attention semantic segmentation network (BDANet). BDANet completes accurate segmentation while reducing parameters, enabling real-time semantic recognition of the railway environment. A COCO-Stuff-Rail dataset extracted based on COCO-Stuff is constructed to guide model training. Then, an adaptive correction algorithm is introduced to fine-tune the BDANet, making it generalizable to diverse realistic environments. Ultimately, this article achieves end-to-end track extraction, open-set foreign object detection, and common foreign object identification using a unified process. To evaluate the superiority of BDANet, comparison, and ablation

experiments are conducted on the COCO-Stuff-Rail. Visual segmentation and open-set detection results of a real-world scenario validate that the proposed process can bridge the gap between the training set and practical applications.

Index Terms—Attention mechanism, boundary-assistant, deep learning (DL), domain adaptation, foreign object intrusion detection (FOID), semantic segmentation.

I. INTRODUCTION

RAILWAYS play a significant role in transportation, providing advantages such as punctuality, comfort, and energy efficiency [1]. As of October 2023, the passenger turnover of China's railroads has reached 1280.259 billion person kilometers, accounting for more than 50% of all means of transportation [2]. Ensuring safety is a fundamental requirement of railway transportation. In light of the characteristics of railway transportation, any intrusion within the railway clearance, known as foreign object intrusion, could result in risks such as train derailment and emergency stops. Consequently, continuous monitoring and protection along the rail line is essential. In fact, with the expansion of the railway network, the potential for widespread delays caused by foreign objects is also increasing [3]. Therefore, reliable foreign object intrusion detection (FOID) is an essential safeguard for driving safety and improving efficiency.

FOID currently in railway systems can be broadly categorized into three types. The first type is passive protection, such as electronic fences and radiation detection, which can only locate the target after being triggered and cannot complete active detection. The second type is train inspection, which is limited to work during the “skylight” time and cannot guarantee continuous monitoring. Besides, manual examination of camera-collected data provides reliable and continuous monitoring but is time-consuming and labor-intensive. To overcome these limitations, artificial intelligence-powered active detection approaches have been extensively researched. However, the environmental diversity caused by the continuously expanding railway network, along with the randomness of foreign objects, poses two main challenges for active detection. One is to enhance the model's generalization, and the other is to detect open-set targets. Therefore, this article proposes an open-set FOID scheme based on semantic segmentation, marking the first attempt to provide a potential solution for the above two challenges.

Received 29 December 2023; revised 9 March 2024; accepted 27 August 2024. Date of publication 16 October 2024; date of current version 18 August 2025. Recommended by Technical Editor L. Cheng and Senior Editor X. Chen. This work was supported in part by the National Natural Science Foundation of China under Grant 62327806, Grant 61925302, and Grant 62273027, in part by the Beijing Municipal Natural Science Foundation under Grant L211021, in part by the Open Project of Shanghai Driverless Train Control System Engineering Technology Research Center under GrantSUTC-2024KT-05. (Corresponding authors: Hairong Dong; Haifeng Song.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TMECH.2024.3468620>.

Digital Object Identifier 10.1109/TMECH.2024.3468620

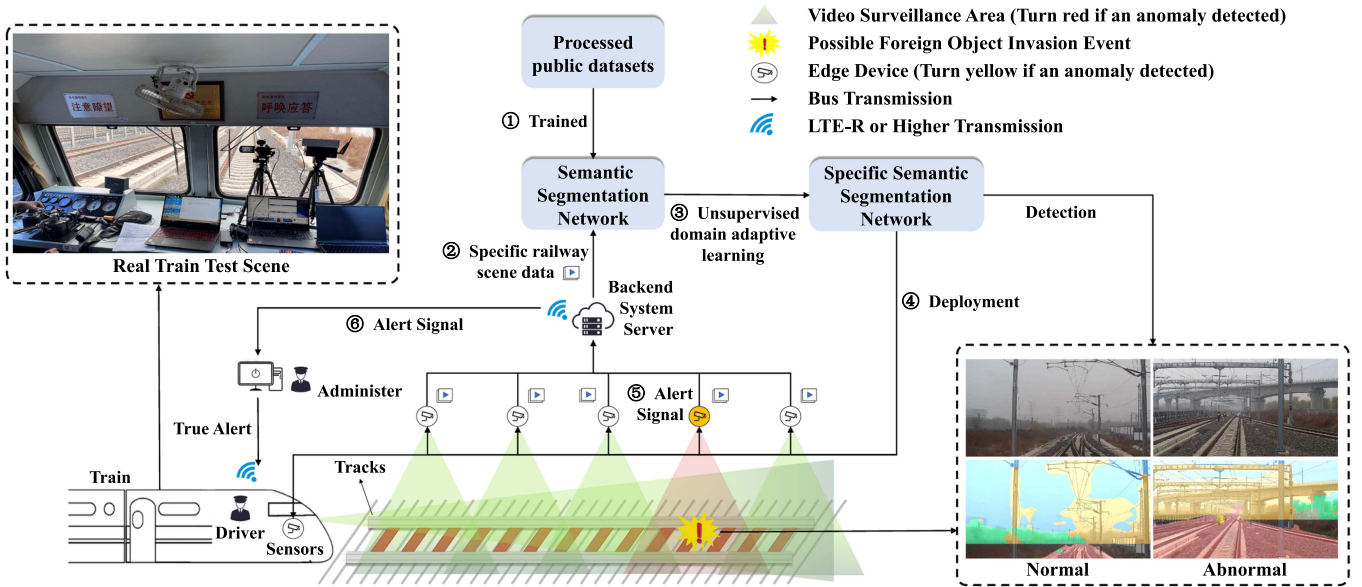


Fig. 1. FOID scheme for the railway. The processed public dataset is COCO-Stuff-Rail. Semantic segmentation network using BDANet and applying MK-MMD [6] as unsupervised domain adaptive method.

A two-stage deep learning (DL) railway FOID scheme considering diverse environments is proposed, and its overall architecture is depicted in Fig. 1. The visual sensors can be onboard or deployed alongside the tracks where foreign objects often occur. If any anomalies are detected, an alert signal will be synchronously transmitted to the driver and the backend administrator for manual confirmation as well as train driving strategy changes [4], [5]. This deployment allows drivers to immediately access the anomalies and respond promptly, avoiding the possibility of inefficiency in cross-layer information processing. The first step of the proposed scheme is to train the boundary-assisted dual-branch attention semantic segmentation network (BDANet) based on a customized dataset COCO-Stuff-Rail. This step ensures the basic segmentation accuracy for railway scenarios. In the second step, a domain adaptive correction algorithm based on multikernel maximum mean square deviation (MK-MMD) [6] is introduced. This algorithm retrains the data captured by a specific visual sensor to ensure the applicability of BDANet to the scenario. This step prevents insufficient model generalization from the limited dataset.

The main contributions of this work include the following.

- 1) To the best of our knowledge, a semantic segmentation-based scheme for railway FOID is proposed for the first time. Considering the simplicity of FOID migration toward different scenarios (onboard or trackside), our scheme combines a pretrained semantic segmentation network and an unsupervised domain adaptive correction algorithm. This integrated approach enables the identification of common foreign objects, including people, trains, animals, stones, and more. Moreover, the scheme has the potential to detect open-set foreign objects. The effectiveness of this scheme is validated through the detection results obtained from realistic onboard views.
- 2) Oriented to the limited storage and computational resources while guaranteeing detection accuracy, a novel

semantic segmentation network, BDANet, is proposed. A global-local attention module (GLAM) is introduced to expand the receptive field and extract representative features, resulting in a lightweight architecture. In addition, boundary information is utilized to improve segmentation accuracy without significantly increasing parameters. We test BDANet on a self-built COCO-Stuff-Rail dataset. BDANet achieves a competitive running speed of 35.83 FPS with 208M of GPU memory.

The rest of this article is organized as follows. Related works on railway FOID and semantic segmentation are discussed in Section II. Section III briefly analyzes the overall architecture of the proposed method and details are introduced in Section IV. The experimental results and the comparison with various state-of-the-art approaches are discussed in Section V. Finally, Section VI concludes this article.

II. RELATED WORK

A. Railway FOID

FOID research can be divided into traditional machine learning algorithms and DL-based target detection algorithms. Traditional approaches typically involve a process consisting of track extraction to identify regions of interest (ROI), succeeded by target detection and identification. Geometric attributes intrinsic to the rail, including rail spacing [7] and parallel tracks [8], often serve as key characteristics for track extraction. With the extracted ROI, background difference [9] and optical flow [10] are employed to detect foreign objects based on pixel changes. Subsequently, machine learning algorithms such as support vector machines (SVM) use manually designed features to identify the detected objects. However, these algorithms are limited in handling complex and dynamic scenarios. Furthermore, distributed processing is not conducive to storage space optimization.

DL has been prominent due to its capacity to automatically extract meaningful features through backpropagation for end-to-end learning [11], [12]. However, the reliability of DL heavily relies on the availability of comprehensive data that accurately represents the desired scenario [13]. Spontaneously, Yu et al. [14] developed a feature extractor based on the faster-RCNN to detect trains, pedestrians, and animals. The extractor is trained on a dataset containing 20 000 images of railway scenarios collected from the Internet. Similarly, Ye et al. [15] customized a dataset of 7342 images depicting railway shunting scenarios recorded under various lighting and weather conditions. Then, a FR-Net is trained on this dataset for identifying trains, track orientation, pedestrians, helmets, and spanners with improved efficiency. The EU Shift2Rail sub-project Smart [16] trained on 998 freight locomotive view images can detect pedestrians, vehicles, bicycle, and large animals based on YOLO.

To implement faster and smaller object detection, Xu et al. [17] proposed a multistage approach to obtain multiscale feature maps to identify objects of different sizes at varying distances. They used a customized dataset comprising 6384 train operation views detecting trains, luggage, and pedestrians. Cao et al. [18] designed a neural network based on depth separable convolution, which is trained on a dataset with 15 036 intrusion images of railway scenarios containing labels of people, cars, rocks, boxes, and dogs. Tian et al. [19] proposed feature fusion CenterNet with variable focus multiscale augmentation (FFCN-VFMS) and trained on a railway clearance intrusion (RCI) dataset. Li et al. [20] introduced a multiblock SSD for detecting small railway targets using drones. He et al. [21] presented an improved method based on R-CNN for FOID, which is trained on a customized dataset including nine types of labeled obstacles.

In conclusion, traditional algorithms are unsuitable for complex and dynamic scenarios. DL-based object detection algorithms require learning the target of interest from the training dataset. As a result, the present research mainly concentrates on detecting prevalent objects like trains, pedestrians, and animals. However, the infrequent occurrence of foreign objects, along with the need to consider model generalization across different lighting conditions and scenarios, complicates the construction of foreign object detection datasets.

B. Semantic Segmentation

Semantic segmentation is a basic and challenging task in computer vision. Thanks to the great success of convolution neural networks (CNNs) in image classification, the fully convolutional network (FCN) [22] has emerged as a leading end-to-end approach for semantic segmentation, replacing fully connected layers with deconvolution. Subsequent studies in CNN-based semantic segmentation have predominantly focused on the drawbacks of FCN. PSPNet [23] and EncNet [24], for instance, respectively, propose a pyramid pooling module and context encoding module to capture more contextual information and expand the receptive field. DeepLabV3+ [25] adopts a spatial pyramid pooling module with atrous convolution to introduce multiscale context. In addition, PointRend [26] proposes an

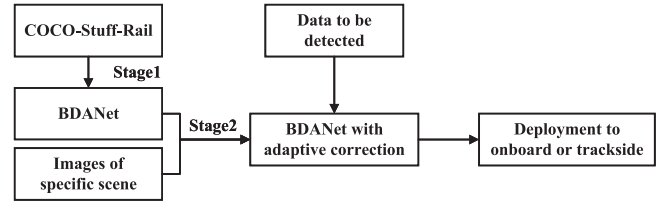


Fig. 2. Block diagram used to outline the implementation of the solution.

iterative subdivision up-sampling algorithm to perform point-based segmentation predictions, improving boundary segmentation. Furthermore, a series of methods aimed at optimizing the convolutional structure and fusing multiscale features have contributed to advancements in semantic segmentation [27], [28], [29].

With the emergence of transformers, semantic segmentation has expanded beyond the limitations of CNN frameworks. Incorporating the self-attention mechanism can effectively capture long-range dependencies between pixels. Therefore, building upon the non-local [30], dual-attention mechanism DANet [31], lightweight attention mechanism GCNet [32], CCNet [33], and others have been gradually developed. To comprehensively capture global contextual relationships, methods like SETR [34] directly replace the encoder CNN with ViT, leveraging the power of vision transformers. SegFormer [35] designs overlapping patches in the encoder to ensure local continuity and eliminate positional embedding while incorporating a lightweight MLP in the decoder. In addition, Segmenter [36] improves feature representation in the decoder by utilizing a series of learnable tokens corresponding to semantic categories. However, the parameters and computational requirements of the Transformer tend to be relatively large, which may limit their suitability for mobile models. Moreover, the performance improvement compared to CNN architectures with small datasets is not always significant.

In contrast to the aforementioned networks, BDANet integrates a global attention module (GAM) after extracting local features using CNN blocks. This integration aims to capture global contextual information and emphasize more representative features. At the same time, local and global features are combined through residual concatenation to form a GLAM. This integration enables the model to capture more contextual information from shallow, coarse-grained features. In addition, the feature selection mechanism employed in GLAM enhances the computational speed and effectively reduces GPU memory without affecting the segmentation accuracy.

III. SCHEME OVERVIEW AND DATA PREPROCESSING

In this section, the implementation process of the proposed solution is outlined, and the creation of the COCO-Stuff-Rail dataset is detailed.

A. Overview of the Scheme

The overall process of the proposed scheme is depicted in Fig. 2. In the first stage, store the initial semantic segmentation model BDANet, pretrained on COCO-Stuff-Rail, in the backend



Fig. 3. Some visual samples of the COCO-Stuff dataset. The first column is partially irrelevant data. The second column is the data with both relevant and irrelevant semantics. Note that the labels have been modified. The third column presents data with incorrect original labeling.

server. The dataset contains samples and possible stuff of the railway scenarios extracted from COCO-Stuff, as described in the following section. The BDANet aims to capture all foreign objects by training it to semantically comprehend the intrinsic elements that would appear in a railway scenario, e.g., railways, trains, buildings, etc. Subsequently, BDANet is retrained using unsupervised domain adaptation based on scenario-specific images in the second stage to ensure generalizability before deploying it to different environments. An adaptive correction algorithm based on MK-MMD [6] is proposed. The unsupervised approach avoids manual labeling of the retrained data, allowing the target domain to utilize raw data without any preprocessing. Simultaneously, the method incorporates a multilayer distribution loss within the network, which aims to minimize the impact of disparate feature distributions between the source and target domains. This process unifies pretraining and retraining within an end-to-end architecture. Consequently, the resulting network is customized for specific scenarios, enabling it to identify common foreign objects and detect open-set ones. By incorporating domain adaptation, the model can be easily deployed in sensor units at any location.

B. Data Preprocessing

The COCO-Stuff dataset [40] is utilized to achieve accurate pixel classification in the railway environment. This dataset includes 164 000 images with semantic annotations covering 172 categories such as “train” and “railroad.” However, it should be noted that the railway scenario samples constitute only a part of it. Including all the data to train the model would inevitably result in blurred focus. Therefore, data preprocessing is performed on the COCO-Stuff to create a subset, referred to as the COCO-Stuff-Rail. Some representative data from COCO-Stuff are shown in Fig. 3. Most of these data are unrelated to the railway, as shown in the first column. These irrelevant

data are excluded from our subset. The second column shows samples where both relevant and irrelevant semantics coexist. For these samples, we retain the relevant semantics and mask the irrelevant ones. Finally, the third column presents a small number of mislabeled samples, which have been corrected. The final generated subset includes 17 distinct categories that may appear in the railway scenarios. Some categories are directly converted from individual labels in COCO-Stuff, while others are defined by merging multiple labels to provide a more generalized description. For example, both “sky-other” and “clouds” are uniformly classified as “sky” to simplify learning. The statistical characteristics of the final filtered subset are shown in Table I. Most of the data are captured under good lighting conditions, with only a few images taken at night. The dataset basically does not involve abnormal weather, such as rain, snow, fog, etc. The model trained based on COCO-Stuff-Rail is capable of detecting and recognizing intruding foreign objects including persons, bikes, cars, stones, and animals.

IV. METHODOLOGY

This article proposed a novel encoder–decoder network called BDANet, which uses boundary cues to optimize the segmentation branch. The overall architecture of BDANet, consisting of two components: GLAM-ResNet as the backbone and dual-branch decoder, is depicted in Fig. 4. Furthermore, an adaptive correction algorithm based on MK-MMD [6] is applied to refine the applicability of BDANet to specific scenarios. Details of these components are described as follows.

A. GLAM-ResNet Encoder

Oriented to practical industrial scenario applications, we propose inserting attention modules into the CNN to obtain enough contextual information for semantic segmentation, avoiding the storage difficulty caused by large parameters and maintaining high detection accuracy. Specifically, a GAM is introduced after each stage of ResNet, and local and global features are fused through residual connections, enhancing the representation. This allows the network to extract global context from the data in the early stages. In addition, Rather than adopting the initial convolution layer in ResNet, we employed dilated convolution layers to expand the receptive field. This particular layer consists of three dilation convolutions with kernel sizes of 3, 5, and 7, all with a dilation rate of 1. The output of this layer is obtained by element-wise summation of the results from these dilated convolutions. As a result, the backbone consists of five stages with a resolution relative to the input of $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{8}$ and $\frac{1}{8}$ to minimize the loss of position information caused by downsampling.

The proposed attention module is shown in Fig. 5. The GAM plays a crucial role in capturing any dependencies between pixels, effectively extending the perceptual field. Specifically, a position and channel dual attention mechanism inspired by the DANet [31] is incorporated. The position attention remains unaltered, the Q, K, and V vectors are obtained through convolution. The matrix product of Q and K enables the calculation of the similarity weight $S_p \in \mathbb{R}^{(H \times W) \times (H \times W)}$ between any two positions, which reflects self-attention. For channel attention,

TABLE I
COUNT OF CATEGORIES PRESENT IN COCO-STUFF-RAIL

COCO-Stuff-Rail (Left) and COCO-Stuff (Right)												
Labels	Sky	106;157	Structural	95;96;102;103; 128;132;151;158; 164;171-177	Person	1	Field	124;163	Bike	2;4	Traffic Light	10
Statistics	20 459	41 694	14 886	146 304	14 223	63 965	8947	23 960	6431	6741	4049	4138
Labels	Water	148;155;178	Plant	97;119;142;169	Plane	5	Car	3;6;8	Stone	162	Sign	13
Statistics	5985	11 364	17 404	59 096	2930	2982	10 970	22 310	1641	1828	1691	
Labels	Mountain	135;127	Ground	111;125;126;140; 144;149;154	Train	7	Railway	125;147	Animal	16-25	All	
Statistics	7206	7385	9046	58 450	3471	3587	3602	3708	6429	25 174	25 839	118 287

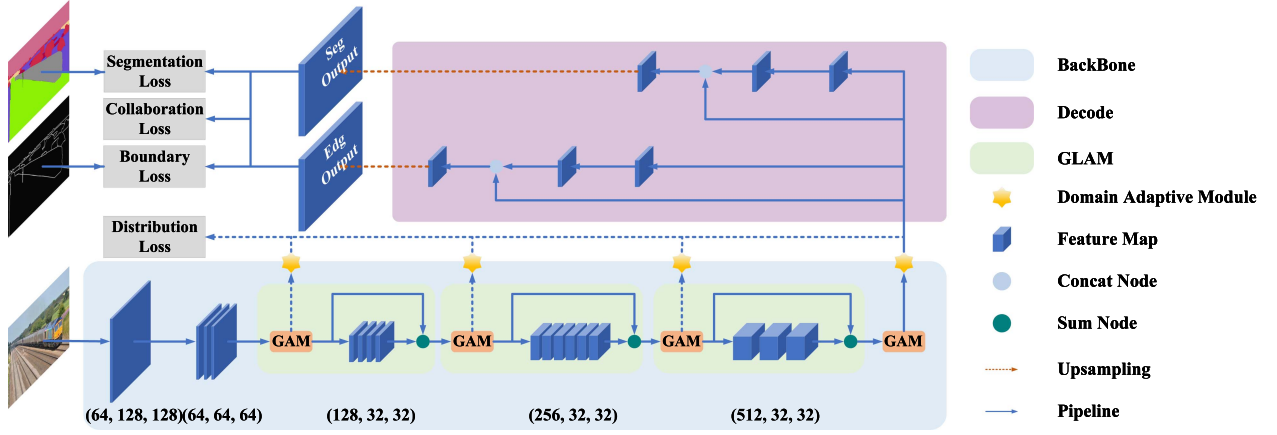


Fig. 4. Overall architecture of BDANet. The domain adaptation module represents the place that computes the statistical distribution discrepancy between the target and source domain.

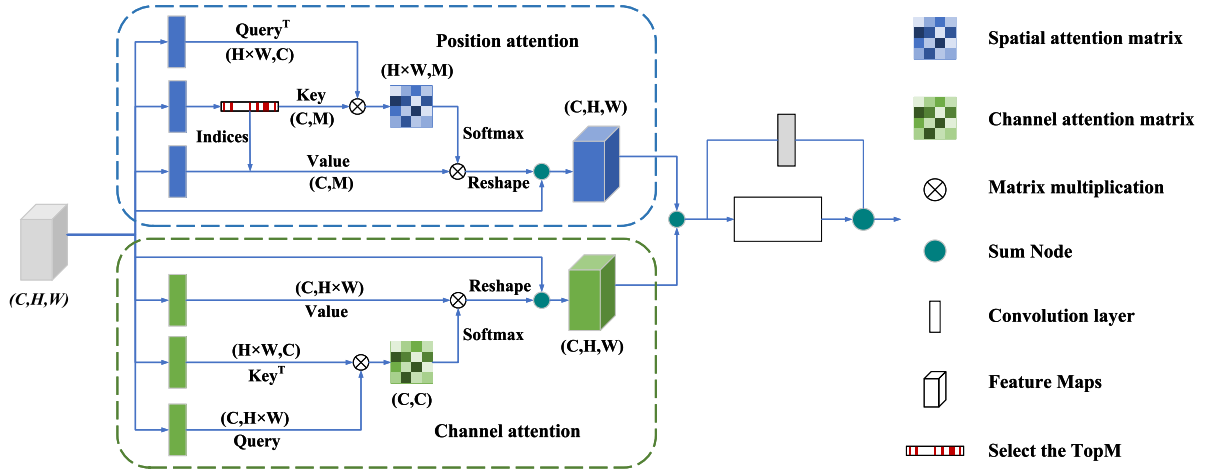


Fig. 5. Overall architecture of GLAM. GAM consists of position self-attention (blue box) and channel self-attention (green box).

we argue that a simple reshaping is insufficient for autonomous weight learning. Therefore, convolution is also employed to transform the feature map into three vectors. However, this process involves square-level computation of the input dimensions and the convolutional channels, resulting in significant computational and storage overheads. Direct processing of images

becomes challenging. This article improves the computation of the spatial attention matrix.

It is widely acknowledged that plenty of low-level semantic features are extracted in the early stages of the model. Consequently, there potentially exists redundant information that can be ignored, and not all dependencies between positions are

worth paying attention to. By selecting the top M maximum correlation values (TopM) in K and their corresponding positions in V , the weight matrix can be transformed from S_p to $S_p^M \in \mathbb{R}^{(H \times W) \times M}$, effectively reducing parameters. Regarding channel attention, it is observed that the ResNet exhibits a fourfold increase in channel output over 50 layers compared to below 50 layers. Therefore, the blocks connected between the two stages are modified, maintaining channel output per stage to be 64, 128, 256, and 512. This modification reduces computational and storage consumption by a factor of 16, enabling the utilization of deeper ResNet for extracting higher-quality features.

B. Dual-Branch Decoder

The accuracy of segmentation heavily relies on precise boundary identification. Considering the simplistic composition and fixed stuff of railway scenarios, a dual-branch decoder combining boundary identification and semantic segmentation is adopted. The learned boundary information is utilized to assist segmentation through a multitasking architecture.

1) *Boundary Learning Branch*: Boundary labels can be easily obtained based on known semantic labels. In this article, the specification of boundary points as pixels that differ from the surrounding labels. It allows for acquiring boundary information for each stuff rather than limiting to the boundary between two stuff. The L_e employs binary cross-entropy loss, incorporating weights w_e^r and w_e^s . w_e^r gives higher weights to samples containing “railway” and “train” in the dataset, while $w_e^s = y_e \cdot N_{\text{neg}}/N_{\text{pos}}$ balances positive and negative samples.

2) *Semantic Segmentation Branch*: A boundary-assisted loss L_d is constructed to optimize the segmentation accuracy as illustrated as follows:

$$L_d = L_{\nu(x)} + L_{\zeta(x)} \quad (1)$$

where $L_{\nu(x)}$ represents the binary cross-entropy loss with a weight w_{ν} , which primarily addresses the category imbalance and increases the weight of “railway” and “train” samples.

$$L_{\zeta(x)} = -\frac{1}{N} \sum_{n=1}^N w_{\zeta} (y_n^g - y_n^b) \log \left(\mathcal{A}(x_n)^g \cdot \left(1 - \mathcal{A}(x_n)^b \right) \right). \quad (2)$$

As described in (2), the network performance is expected to be optimized by $L_{\zeta(x)}$, which aims to incorporate the output of the boundary learning branch into the segmentation branch by finding potential relationships between boundary and each stuff. In (2), $\mathcal{A}$, x_n and y_n denote BDANet, input image and label, respectively. w_{ζ} is same with the category balance weight in w_{ν} . The superscripts g and b denote the semantic segmentation branch and the boundary learning branch, respectively. It is known that the closed regions composed of boundaries all belong to the same label. Based on this prior, $L_{\zeta(x)}$ takes the semantic information excluding the boundary as the ground truth. When the boundary result and the segmentation result converge to the ground truth infinitely, the joint loss will also converge to zero infinitely.

C. Adaptive Correction Algorithm

The adaptive correction algorithm consists of an unsupervised domain adaptive module and a postcorrection algorithm, as described in Algorithm 1. The unsupervised domain adaptation module adopts the MK-MMD [6] for its simplicity without more parameters or new models. By connecting it after each GLAM, this module extracts feature from every stage of the backbone, calculating the distribution discrepancy between the source and target domains as a loss to guide network adaptation to the target domain. The distribution discrepancy is formulated as follows:

$$L_{\Phi(s,t)} = \frac{1}{k} \sum_{i=1}^k \left\| \frac{1}{n} \sum_{j=1}^n K_i(s_j) - \frac{1}{m} \sum_{j=1}^m K_i(t_j) \right\|_{\mathcal{H}}^2 \quad (3)$$

where s, t denote the source and target domain samples, respectively, mapped to Hilbert space by different kernel function K to compute the expectation gap. k represents the number of kernel functions employed.

The postcorrection algorithm is applied to refine the model’s predicted results. This is due to the inherent uncertainty of the network, which could lead to high false positives when directly applied to detection. The postcorrection algorithm first identifies the ROI. Specifically, it employs Hough transformation to locate two lines with centers closest to the railway’s edge pixels in the segmentation result y^g as the boundaries of the track. The intersections extended from these lines pass through sky-track demarcation line. The ROI is determined as the region within the track and above the demarcation line. Then, the detected foreign objects in the y^g are re-evaluated to enhance the reliability of the detection.

V. EXPERIMENTAL RESULTS

A. Datasets and Evaluation Metrics

1) *COCO-Stuff-Rail*: This dataset processed by ourselves provides pixel-level semantic annotations for railway scenarios related to “things” and “stuff,” with a resolution of 256×256 . It includes 23 255 and 2584 images respectively for training and validation, as well as 2728 test images. The dataset includes 17 distinct categories that may appear in the railway scenarios. 2) *Evaluation Metric*: The following measures are used to evaluate the performance. 1) Average intersection union (mIoU) used to evaluate segmentation accuracy. 2) GFLOPs, occupied GPU memory, the number of network parameters, and FPS are used to evaluate the running efficiency.

B. Training Setting

BDANet is implemented on a hardware platform equipped with dual RTX 4090 GPUs, dual Intel Xeon Gold 8160 2.1 GHz CPUs, 64 GB of memory, and the Windows 10 operating system. All models are constructed based on Pytorch 2.0 using Cuda 11.8 and trained using the adaptive moment estimation (Adam) optimization algorithm with batch size 16. The initial learning rate for GLAM-ResNet is 1×10^{-4} , and for others is 1×10^{-3} . The dynamic learning rate adjustment parameters are set to $\gamma = 0.5$ and $\text{step} = 30$, respectively. Any data augmentation operations are not used. The implementation for

Algorithm 1: The Adaptive Correction Algorithm.

Input: source domain data s ; target domain data t
Output: Corrected segmentation predictions y^β

```

1: for  $epoch = 1, 2, \dots$  do
2:    $\mathcal{B} = \mathcal{A}(s, t, L_e + L_d + \lambda L_{\Phi(s,t)}); \triangleright$  update weights
3: end for
4:  $y = \mathcal{B}(x); \triangleright$  predicted results
5: Find ROI  $y^\alpha$ ;
6: Find all closed region  $\Upsilon^\alpha$  based on  $y^\alpha$ ;
7: for  $\Upsilon_i^\alpha, i = 1, 2, \dots$  do
8:   if  $y_i^\alpha$  is label of unknown then  $\triangleright$  calculate similarity
9:     for  $\Theta_j^\alpha$  close to  $\Upsilon_i^\alpha, j = 1, 2, \dots$  do
10:       $\delta = \frac{1}{2} \left( \sum x_i \log \left( \frac{x_i}{\frac{x_i + x_j}{2}} \right) + \sum x_j \log \left( \frac{x_j}{\frac{x_i + x_j}{2}} \right) \right);$ 
11:    end for
12:    if  $\min(\delta) < threshold$  then
13:       $y_i^\alpha = y_j^\alpha;$ 
14:    end if
15:  end if
16: end for
17: Obtain revised predictions  $y^\beta$ , find all closed region  $\Upsilon^\beta$  based on  $y^\beta$ ;
18: for  $\Upsilon_i^\beta, i = 1, 2, \dots$  do
19:   if  $y_i^\beta$  is label of foreign objects then
20:     for  $\Theta_j^\beta$  close to  $\Upsilon_i^\beta, j = 1, 2, \dots$  do
21:       calculate similarity;
22:     end for
23:     if  $\min(\delta) < threshold$  then
24:        $y_i^\beta = y_j^\beta;$ 
25:     end if
26:   end if
27: end for
```

all selected baselines is based on an open-source project MMsegmentation [41] for semantic segmentation. Due to the modified backbone of BDANet, GLAM-ResNet50 is first pretrained in the whole ImageNet dataset for 60 epochs. Considering both efficiency and fairness, all other baselines and BDANet in the comparison study are conducted with 80 epochs, using a training set of 4500 images and a validation set of 500 images from COCO-Stuff-Rail. In the adaptive correction algorithm study, the first stage of BDANet pretraining is based on the whole COCO-Stuff-Rail. The mentioned first stage is described in Fig. 2.

C. Comparisons With State-of-the-Arts

Some baselines are selected for comparison to validate the effectiveness of BDANet, as shown in Table III. For a fair comparison, the resolution of input images is 256×256 in all experiments, and the backbone is mainly ResNet50. It is worth noting that ResNet101 is not employed due to the small training set with 4500. Without additional data, the performance of ResNet101 does not differ much from that of ResNet50. In Table III, it can be found that BDANet outperforms most of the previously mentioned networks [26], [31], [33], [37], achieving 74.4% mIoU. Particularly, compared to networks

that also employ boundary information, such as PIDNet [38] and OCRNet [39], BDANet exhibits improvements of 3.5% and 1.0% in mIoU, respectively. Among the baselines, Segmester [36] achieves the highest performance with a mIoU score of 76.5%. This is attributed to its utilization of the Transformer architecture with more model parameters. Furthermore, Table II reports the comparative results for each category. It is observed that our approach achieves the highest IoU scores in eight categories. Compared to the best-performing baseline, our method improves the IoU of “ground,” “stone,” and “train” by 1.1%, 0.9%, and 1.4%, respectively. It can be seen that although the accuracy of BDANet is not the best due to the fewer parameters, it is not too far off.

D. Ablation Study

1) Ablation Study of Different Components in BDANet:

Table V presents some ablation studies on the validation set, which quantify the impact of two main components: GLAM ($w/o L$) and dual-branch, respectively. The experiment indicates that each component is continuously improving performance. Specifically, GLAM-ResNet with L achieves 73.2% mIoU, which brings a 4.7% enhancement while reducing the parameters by 15.8M compared to the baseline ResNet50. It shows that GLAM plays an important role in expanding the shallow network receptive field. Precisely because the GLAM expands the receptive field, the effectiveness of the boundary-assisted branch is significantly improved. Define the effectiveness mentioned here as the ratio of the mIoU growth rate to the parameter growth rate. As seen in rows 5 and 6 of Table V, without GLAM, the dual-branch architecture only increased the mIoU by 0.2%, while adding 23.6M parameters. Upon observing rows 2 and 4, the adoption of the dual-branch architecture increased the mIoU by 1.2%, with only a 1.5M increase in parameters, improving effectiveness by approximately 33.6%. Ultimately, combining both components yields a remarkable 5.9% improvement in mIoU, accompanied by a reduction of 14.3M parameters. Moreover, when training on a dataset of 4500 samples, the performance of using a deeper network, such as GLAM-ResNet101, is comparable to that of GLAM-ResNet50 due to the limited data. However, if trained on the complete COCO-Stuff-Rail dataset, the deeper backbone yields a higher mIoU. In summary, the introduction of GLAM effectively improves the segmentation accuracy while reducing the parameters. The dual-branch decoder also plays an important role in accuracy improvement while avoiding significant increases in parameters.

2) *Ablation Study for M in GAM:* This section evaluates the effect of M on the mIoU, GFLOPs, and GPU memory in the TopM process. Table VI presents the variations for different $r = N/M$, where N represents the product of the input image’s width and height in pixels. It is observed that the segmentation accuracy peaks when $r = 16/16/1/1$. The four values denote r in each GAM. Observing rows 1, 2, 3, 5, 7, and 9 in the Table VI, when the two GAMs in the shallow layers adopt larger r , the mIoU initially decreases and then increases. As r continues to increase, the mIoU again shows a downward trend. The 0.1% change in mIoU in rows 2 and 3 is likely influenced by training randomness and does not indicate a trend. This indicates that

TABLE II
INDIVIDUAL CATEGORY RESULTS AND THE AVERAGE OVER ALL CATEGORIES OF THE COCO-STUFF-RAIL VALIDATION SET IN TERMS OF mIoU SCORES

Method	sky	structural	plane	field	person	car	mountain	plant	ground	sign	stone	water	train	animal	bike	railway	trafficlight	mIoU (%)
Semantic FPN [37]	91.9	66.8	77.8	72.5	73.7	69.0	61.8	68.0	69.7	87.4	37.9	79.7	79.0	76.8	82.6	62.5	41.2	70.5
PIDNet [†] [38]	92.4	67.5	76.3	74.0	70.8	64.3	60.8	68.8	71.7	88.5	51.8	82.6	80.2	75.9	77.4	62.7	39.3	70.9
PointRend [26]	91.9	67.9	77.3	72.2	73.1	66.5	60.8	67.5	70.5	90.9	52.0	79.9	79.0	76.3	81.5	63.8	41.2	71.3
DANet [31]	92.3	67.7	73.4	74.5	75.9	65.3	62.1	68.3	73.3	91.5	50.5	83.1	76.1	79.1	82.1	63.6	47.1	72.1
ENCNet [24]	92.1	68.0	76.9	74.4	74.5	67.6	63.3	68.5	70.5	89.8	52.0	82.5	80.2	77.5	82.6	64.1	45.2	72.3
DeepLabV3+ [25]	92.8	69.8	79.3	75.1	74.8	69.3	63.4	69.1	71.0	92.7	43.0	82.7	80.0	78.2	83.2	63.5	44.8	72.5
OCRNet [†] [39]	92.3	69.5	75.7	75.9	75.9	69.6	65.1	70.7	71.0	91.6	51.0	81.8	79.7	79.0	84.5	66.2	49.1	73.4
PSPNet [23]	92.5	69.6	79.9	75.0	75.6	69.4	65.0	68.9	73.3	92.0	49.9	82.7	81.0	77.9	84.5	66.1	45.9	73.5
CCNet [33]	92.5	69.5	79.1	75.4	75.8	66.9	64.6	69.9	71.1	91.4	57.5	83.7	79.0	78.2	84.5	64.8	50.1	73.8
UperNet [40]	92.8	70.1	81.0	74.8	77.6	69.7	63.9	70.2	71.8	94.2	55.9	80.3	80.4	81.1	84.1	63.3	48.3	74.1
Ours	93.2	70.3	81.8	73.5	74.7	69.3	66.3	70.8	74.4	93.1	58.4	83.5	82.4	76.0	83.6	65.5	48.6	74.4

The best performance for each individual class is marked with a bold-face number.

TABLE III
EVALUATION RESULTS OF BDANET AND OTHER STATE-OF-THE-ARTS ON COCO-STUFF-RAIL VALIDATION SET

Method	Year	Backbone	mIoU(%)
STDC [42]	CVPR2021	STDC2	64.1
K-Net [43]	NeurIPS2021	ResNet50	65.4
DDRNet [44]	T-ITS2022	DDRNet	68.3
Semantic FPN [37]	CVPR2019	ResNet50	70.5
PIDNet [38]	CVPR2023	PIDNet	70.9
PointRend [26]	CVPR2020	ResNet50	71.3
DANet [31]	CVPR2019	ResNet50	72.1
OCRNet [39]	ECCV2020	ResNet50	73.4
CCNet [33]	ICCV2019	ResNet50	73.8
Segmenter [36]	ICCV2021	ViT-Base	76.5
ours	-	GLAM-ResNet50	74.4

not all positional dependencies in shallow features are worth attention. In addition, comparing rows 2 and 4, 5 and 6, as well as 7 and 8, it is evident that changing r in deeper GAMs typically results in performance degradation. This highlights the importance of considering all semantic correlations between deep features. Furthermore, changes in r have minimal impact on GFLOPs. The effect of r on GPU memory is evaluated by calculating it across the entire validation set when the batch size is 8. It can be found that reducing r to 4/4/1/1 results in no further change, effectively reducing GPU memory consumption by 1.1G compared to the highest 3.3G. This reduction in storage consumption highlights the effectiveness of the attention mechanism. The experimental results show that there is indeed redundant information in low-level semantic features that do not need to be computed. Allowing the model to learn noteworthy features can reduce computational and storage resources.

3) *Ablation Study on Implementing Efficiency*: To analyze the running efficiency of BDANet, we compared it with some baselines in terms of mIoU, model parameters, GPU memory, GFLOPs, and FPS. In all experiments, the images are input to the network with a resolution of 256×256 and a batch size of 8. As depicted in Table IV, BDANet outperforms the selected baselines with the fastest running speed (35.83 FPS) and smaller GPU memory usage (284MB), while utilizing only 32.8M model parameters, which is less than most of the baselines. Although BDANet's mIoU is slightly lower than that of Segmenter, the reduction in model parameters and GPU memory is significant. Considering the achieved FPS and the calculation of redundant time, even when the train operates at 300 km/h, detection can be

performed approximately every 3 m, as long as clear images are captured. This deployment allows drivers to receive immediate alerts of anomalies and respond promptly, avoiding inefficiencies in cross-layer information processing.

E. Visual Results With Adaptive Correction Algorithm

As shown in Fig. 7, the real-scenario data is captured from a forward view of train operation at a circular railway test base with a total length of 38 km, where trains operate at 80 Km/h. The resolution of the captured video is 1920×1080 . Initially, for unsupervised domain adaptation, 6490 and 750 images are extracted from COCO-Stuff-Rail containing “railway,” “train,” and “traffic light” elements as the training set and validation set, respectively. The target domain also involved collecting 6490 images from the video, which are input synchronously to update the parameters of BDANet. The pre-training parameters are obtained by training the BDANet on a training set of 23 255 images and a test set of 2584 images divided by the COCO-Stuff-Rail. The mIoU achieved with this parameter is improved to 78.6% compared to 74.4% trained on 4500 images. Partial visualization results are shown in Fig. 6. Without the adaptive correction algorithm, misidentified “train” are prone to occur in segmentation results. After correction, the performance significantly improved. The results indicate that a simple unsupervised domain adaptation method can quickly adjust the network to adapt to the desired railway scenarios. This phenomenon can be attributed to the customization of the foundational training dataset for railway scenes. This specialization enables the network to potentially extract and adapt similar features more easily. On the other hand, it may also be due to railway scenes exhibiting more regularity. It is worth noting that the proposed adaptive correction algorithm yields better results when applied to clear images obtained under favorable lighting conditions. The essence of the proposed generalization method lies in reducing the distribution gap between the source domain and the target domain, enabling the model to adapt to the target domain's feature space for classification. Therefore, the richness of the data encompassed in the source domain dataset plays a decisive role in determining the model's robustness. Currently, the customized dataset is sourced from COCO-Stuff, consisting of just over 3500 complete railway scene images, which is relatively limited in terms of variations in weather conditions or lighting. Applying transfer learning to real captured images

TABLE IV
COMPARISON OF IMPLEMENTING EFFICIENCY ON COCO-STUFF-RAIL VALIDATION SET IN TERMS OF mIoU, MODEL SIZE, GPU MEMORY, AND GFLOPS

Method	Backbone	mIoU(%) $\uparrow$	Paras(M) $\downarrow$	Mem(M) $\downarrow$	FLOPs(G) $\downarrow$	FPS $\uparrow$
K-Net [43]	ResNet50	65.4	59.9	887	48	11.15
DDRNet [44]	DDRNet	68.4	20.3	368	4	17.78
Semantic FPN [37]	ResNet50	70.5	28.5	481	11	15.72
PIDNet [38]	PIDNet	70.9	37.3	566	9	17.58
PointRend [26]	ResNet50	71.3	28.7	500	12	11.91
STDC [42]	STDC	64.1	12.3	281	3	14.64
DANet [31]	ResNet50	72.1	66.5	938	70	8.93
EncNet [24]	ResNet50	72.3	33.5	591	35	19.75
OCRNet [39]	ResNet50	73.4	36.5	597	123	14.96
CCNet [33]	ResNet50	73.8	47.4	719	50	14.24
UPerNet [40]	ResNet50	74.1	64.1	887	59	14.33
Segmenter [36]	ViT-Base	76.5	102.0	1202	28	12.03
Ours	GLAM-ResNet50	74.4	32.8	284	35	35.83

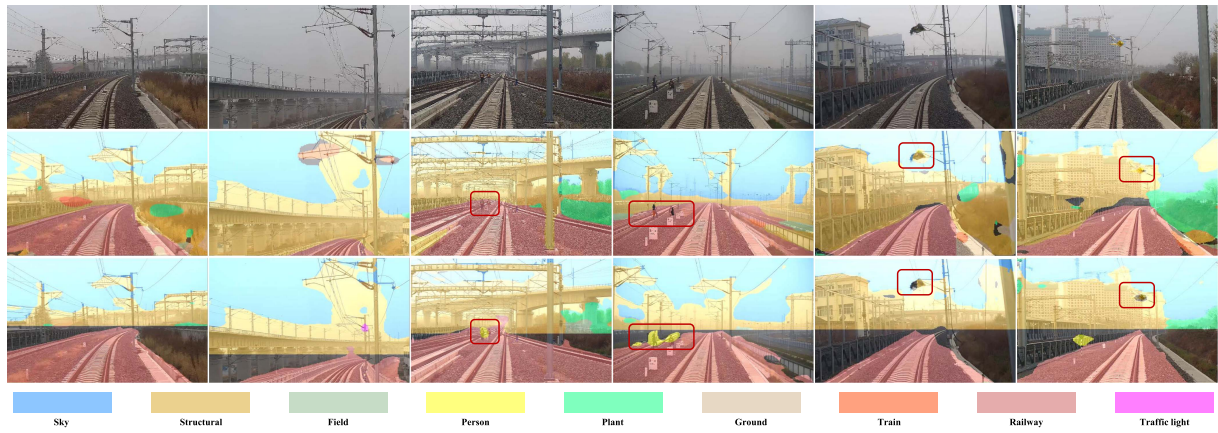


Fig. 6. Visualization results: From top to bottom are the input images, followed by semantic segmentation results without and with the adaptive correction algorithm. From left to right, the third and fourth columns show the foreign object identification results, while the fifth and sixth columns present the open-set foreign object detection results.

TABLE V
CONTRIBUTIONS OF GLAM AND DUAL-BRANCH ENCODER IN BDANET

Backbone	GLAM		Dual-branch	mIoU(%) $\uparrow$	Paras(M) $\downarrow$	FLOPs(G) $\downarrow$
	w/o L	w/ L				
Res101		✓	✓	74.4	51.8	54.4
Res50		✓	✓	74.4	32.8	34.9
Res50	✓		✓	74.2	32.6	34.7
Res50		✓		73.2	31.3	33.4
Res50			✓	68.7	70.7	73.3
Res50				68.5	47.1	49.1

The bold values represent the best performance for each metric under different conditions.

TABLE VI
EXPERIMENTS ON THE IMPACT OF VARIETY R ON mIoU, GFLOPS AND GPU MEMORY

r	mIoU(%) $\uparrow$	FLOPs(G) $\downarrow$	Mem(G) $\downarrow$
1/1/1/1	74.2	34.92	3.3
2/2/1/1	73.1	34.89	2.6
4/4/1/1	73.2	34.88	2.4
4/4/2/1	73.0	34.87	2.4
8/8/1/1	74.3	34.87	2.4
8/8/2/1	72.1	34.87	2.4
16/16/1/1	74.4	34.87	2.4
16/16/2/1	73.2	34.86	2.4
32/32/1/1	73.4	34.86	2.4

The bold values represent the best performance for each metric under different conditions.



Fig. 7. Data acquisition platform in measurement site.

and detecting foreign objects with considerable accuracy poses challenges.

In addition, to verify the detection and identification of common intrusions, we selected 200 images containing targets like people and vehicles, and statistically assessed the ability of the proposed process as shown in Table VII. The results of visualization are shown in the third and fourth columns from left

TABLE VII
FOREIGN OBJECT DETECTION RESULTS WITH AND WITHOUT ADAPTIVE CORRECTION ALGORITHM

	TP/TN	FP	FN
w/o	75	94	31
w	154	43	3

TP/TN denotes correct detection, FP denotes missed detection, and FN denotes False detection.

to right in Fig. 6. It is apparent that the model without correction is more prone to missing detections and producing false alarms, exhibiting low accuracy. The corrected algorithm improves it more than twice. Meanwhile, to examine whether the proposed method can detect open-set foreign objects, images merging with pictures of searched foreign objects on the Internet are input for detection. The results are depicted in the fifth and sixth columns from left to right of Fig. 6. we tried to detect unknown foreign objects such as floating objects on the catenary and toolboxes that may be left on the track. Whether floating objects can be detected mainly depends on their material and shape. If the floating object cannot be clearly distinguished from traffic lights and buildings, there is a propensity for misidentification. The toolboxes can be detected most of the time and occasionally confused with stones. This shows the potential of the proposed scheme to detect open-set foreign objects.

VI. CONCLUSION

This article proposes a scheme based on semantic segmentation for railway FOID for the first time. The scheme trains the semantic segmentation network on the customized COCO-Stuff-Rail dataset, enabling the network to identify common foreign objects and indirectly detect open-set foreign objects that have not been encountered during training. Hence, a novel encoder-decoder network, BDANet, is designed. The encoder of BDANet adopts GLAM to capture global context and extract representative features, resulting in lightweight architecture without compromising precision. The decoder in BDANet introduces a boundary learning branch to improve the network's discrimination of boundary points. Furthermore, considering the multiscenario generalization of the model, a corrective algorithm based on a domain adaptive theory is proposed. The algorithm is directly integrated with BDANet to ensure that pretraining and retraining are unified within an end-to-end architecture. Experimental results confirm that BDANet exhibits superior real-time performance while consuming fewer hardware resources to meet onboard foreign object detection. The visualization results demonstrate that the adaptive correction algorithm enhances the BDANet's generalizability across diverse scenarios by unsupervised retraining the model on images from specific scenarios.

While our work demonstrates great potential in railway FOID, it does have certain limitations. First, it relies on clear image capture, and the effects of varying weather conditions have not been considered. Second, iterative optimization methods for domain adaptation have yet to be explored. Finally, the detection of open-set foreign objects relies on distinct discriminative features, which are prone to confusion. In future research, we

will continue to optimize our proposed scheme from these three areas to facilitate practical deployment in industrial settings.

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